# Arunabha Majumder

**Robotics Mechanical Engineer**

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## EDUCATION

**PhD, Mechatronics and Robotics**
Aalborg University | 2023 - present
Thesis expected March

**MTech, Mechatronics**
AcSIR - CSIR-CMERI, India
2020 - 2022

**BTech, Mechanical Engineering**
Siddaganga Institute of Technology, India
2015 - 2019

**Senior Diploma in Fine Arts (Painting)**
Bangiya Sangeet Parishad
2008, First Division with Distinction

## STUDY ABROAD

**National University of Singapore**
Sep - Nov 2025 (during my PhD)

## SKILLS

**CAD:** SolidWorks (CSWA), technical drawings, stiffness analysis in SolidWorks, Fusion 360 (basic, last used 2019)

**Mechanisms:** kinematic, torque and force calculations, Jacobian-based force estimation, worm gears, springs, GA optimisation

**Build and test:** 3D printing, machining (Bachelor's), assembly, test setups, motors, encoders, force/torque sensors, Arduino, MATLAB/Simulink, LabVIEW

**Art and design:** hand sketching, painting, Adobe Illustrator, Photoshop, Procreate

## ACHIEVEMENTS

Best Paper Award, IFToMM ISRM 2026

Best Major Project Award, 2019

TATA Technologies Prayatna grant

GATE qualified

## LANGUAGES

English (C1), Bengali (native), Hindi, German (willing to learn)

## REFERENCES

**Prof. Shaoping Bai**
Aalborg University, sbh@mp.aau.dk

**Assoc. Prof. Lars Rosgaard Jensen**
Head of Section and Head of Labs, AAU

**Assoc. Prof. Anderson de Souza Castelo Oliveira**
Materials and Production, AAU

## PROFILE

I am a mechanical engineer who designs robot mechanisms, actuators and electromechanical assemblies, and then builds and tests them. Over the last 6 years I have worked on exoskeletons, a 5-bar parallel robot and a soft exosuit: from statics, kinematics and torque calculations to SolidWorks CAD, 3D printed prototypes, manufactured parts and tests on real hardware, always working closely with electrical and software engineers.

## EXPERIENCE

### Research Assistant

Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid shoulder exoskeleton: a variable stiffness mechanism (spring pretensioned by a worm gear) working in parallel with a 6 Nm motor; together they give the torque needed to lift the arm. It uses over 25% less motor energy (Best Paper Award, IFToMM ISRM 2026).
- Optimised the mechanism with a genetic algorithm (diameter, spring stiffness, pretension force), made the CAD and drawings in SolidWorks, 3D printed the first parts and had the final parts manufactured.
- Integrated the motor, encoder and force/torque sensors and measured its stiffness on a physical setup.
- Designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM); estimated its stiffness with less than 10% error and quantified human arm impedance. Force estimation used the variable stiffness actuators and the robot's Jacobian.
- Arm impedance study with the 5-bar robot (6 participants): Qualisys cameras tracked the arm, and I implemented an Arduino-based trigger to sync the robot's motion with the recording.
- Taught university courses and supervised Bachelor's students (2023 - 2025).

### MTech Researcher, Soft Exosuit

CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic artificial muscles for a soft exosuit. Designed the outer PID loop, with integral anti-windup and error nonlinearisation, of a neural network-based gain-scheduled controller in MATLAB/Simulink and implemented it in LabVIEW to drive the muscle (0.3 - 0.78% error).

### Bachelor's Projects

Siddaganga Institute of Technology, India | *2017 - 2019*

- Built an arm exoskeleton and exo glove for a paralysed arm: CAD, machining, assembly and testing.
- Built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018) with student teams.

## PUBLICATIONS

- Hybrid actuated shoulder exoskeleton, IFToMM ISRM 2026 (Springer), Best Paper Award.
- Neural network-based control of a pneumatic artificial muscle, IEEE CONECCT 2022.

## PORTFOLIO

Videos and results from my work: arunabha22.github.io/arunabha-majumder